

Ben-Gurion University of the Negev

Electrical & Computer Engineering

Advanced CPU Architecture & Hardware Accelerators — Course 361-1-4693

FINAL PROJECT REPORT

Structural RV32IM Micro-Controller Unit (MCU)

GPIO · Basic Timer · Interrupt Controller · UART · Restoring-Divider Accelerator

Intel Cyclone V — DE10-Standard (5CSXFC6D6F31C6)

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1. Aim and Assignment Definition

This report documents the design, synthesis, implementation and functional verification of a complete RV32IM micro-controller unit (MCU) on an Intel Cyclone V FPGA. The MCU is built around the scalar single-cycle RV32IM core developed in LAB5 and extends it into a full micro-controller by adding a memory-mapped peripheral subsystem and an interrupt mechanism.

The design is delivered in incremental configurations, each of which is characterised independently for Power, Performance and Area (PPA):

- MCU with GPIO — the RV32IM core plus the memory-mapped GPIO block (switches SW, LEDs LEDR, six seven-segment displays HEX0–HEX5) and the Basic Timer, connected over the shared data bus.
- MCU with GPIO and Interrupt Capability — the above plus the Interrupt Controller (IE / IFG / IR / Type registers, GIE, INTA hand-shake and vector generation), the UART transceiver and the restoring-divider hardware accelerator with its clock-domain-crossing wrapper.
- Pipelined MCU with GPIO and Interrupt Capability — the same peripheral subsystem attached to the 5-stage pipelined RV32IM core of LAB5 Part 2.

All blocks are described in structural/behavioural VHDL and instantiated in a structural top level (MCU_top.vhd). The system keeps the Harvard organisation of the core — separate ITCM and DTCM tightly-coupled memories of 8 kB each — while all peripherals are mapped into the DTCM address space and share the bidirectional 32-bit DataBus. A PLL derives the CPU clock MCLK from the 50 MHz board clock; a second PLL output (DIVCLK) drives the divider accelerator in its own faster clock domain.

1.1 Memory-mapped peripheral interface

Every peripheral follows the same bus protocol: an address decoder produces a chip-select (CS_w), MemWrite_i latches the bus into the addressed register, and MemRead_i together with CS_w enables a tri-state driver (BidirPinDataBus) that returns the selected register onto DataBus. Because only one enable can be active at a time, the peripherals arbitrate the shared bus without any additional glue logic. The CPU therefore accesses the timer, GPIO, interrupt controller, UART and divider with ordinary lw/sw instructions.

1.2 Interrupt mechanism

Interrupt sources (the timer BTIFG, the GPIO push-buttons KEY1–KEY3 and the UART) are captured in the IFG register, masked by IE and by the global GIE bit, and combined into a single INTR request towards the core. The core answers with INTA; the controller then places the corresponding vector taken from the Type register onto the bus, and the ISR FSM inside CONTROL.VHD redirects the PC to the service routine and back. KEY0 is reserved as the asynchronous system reset.

2. RTL Viewer Results

The Quartus RTL Viewer confirms the structural hierarchy of each block. The figures below show the elaborated netlists of the four main peripherals; in each of them the address-decode multiplexer chain is visible on the left, the memory-mapped register bank in the middle, and the read-back multiplexer with the tri-state bus driver (BidirPinDataBus) on the right.

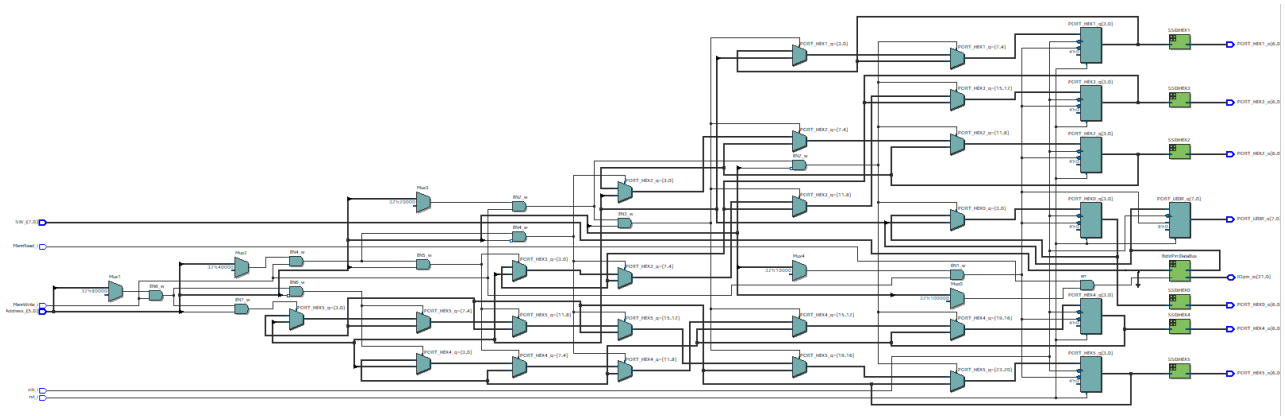


Figure 1: GPIO — RTL Viewer. Address decoding (Mux0–Mux4) drives the enable signals $EN1_w$ – $EN7_w$ of the LEDR/HEX0–HEX5 port registers; each HEX register feeds an SSD decoder before leaving the block.

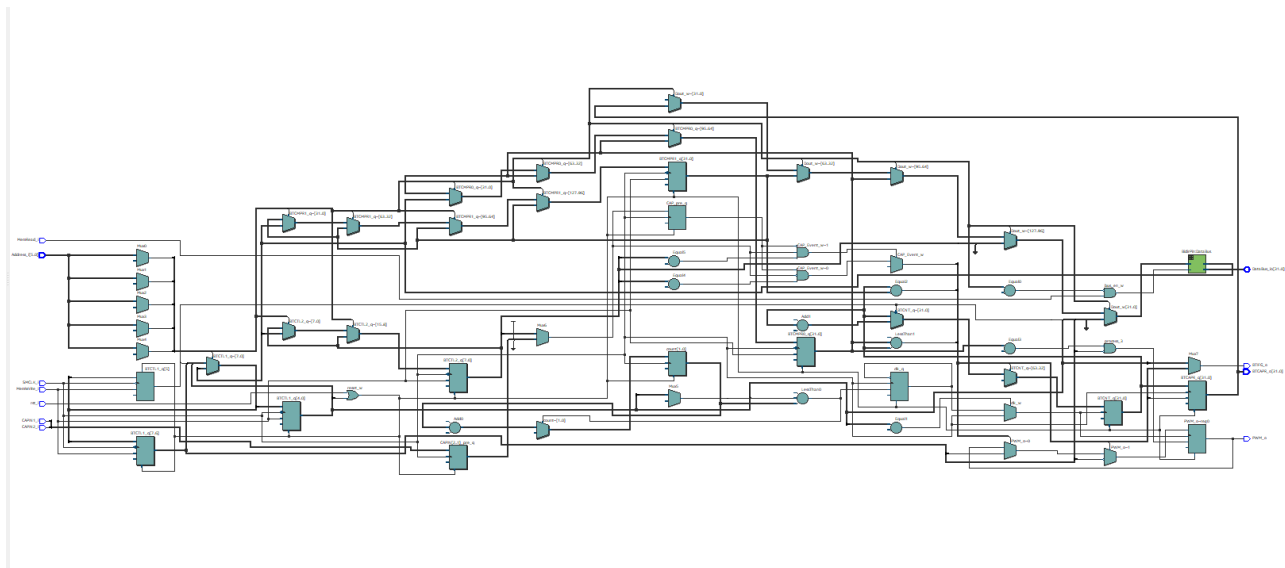


Figure 2: Basic Timer — RTL Viewer. BTCTL1/BTCTL2, BTCMPR0/BTCMPR1 and BTCAPR registers, the clock-divisor chain (clk_q / clk_w), the BTCNT up-counter with its EQU0/EQU1 comparators, the capture logic (CAP_pre_q , CAP_Event_w) and the PWM output generator.

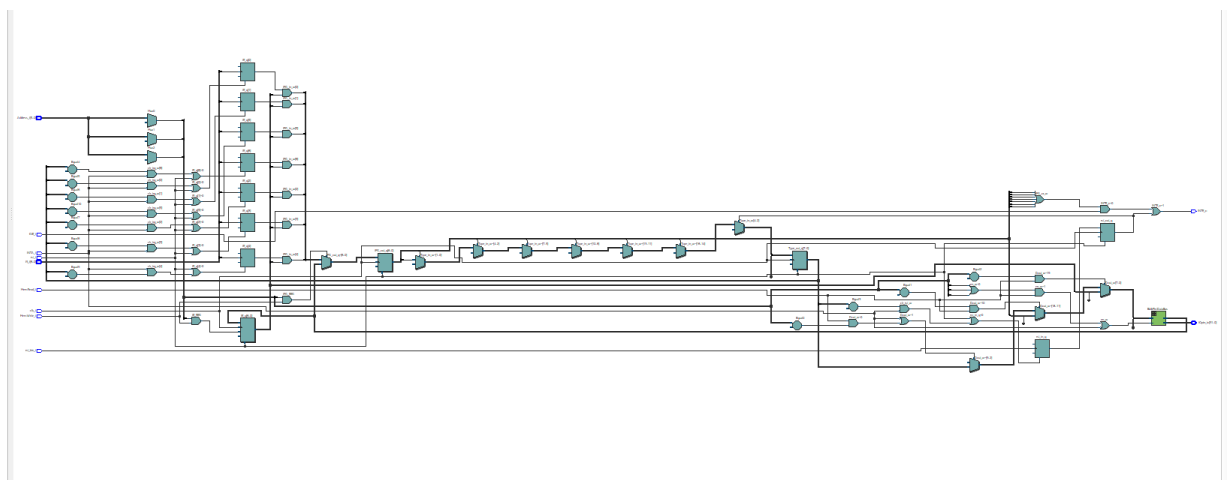


Figure 3: Interrupt Controller — RTL Viewer. The IE / IFG / IR / Type register bank on the left, the priority-encoded vector generation and the Type_out pipeline in the middle, and the INTR output logic plus the bus read-back path on the right.

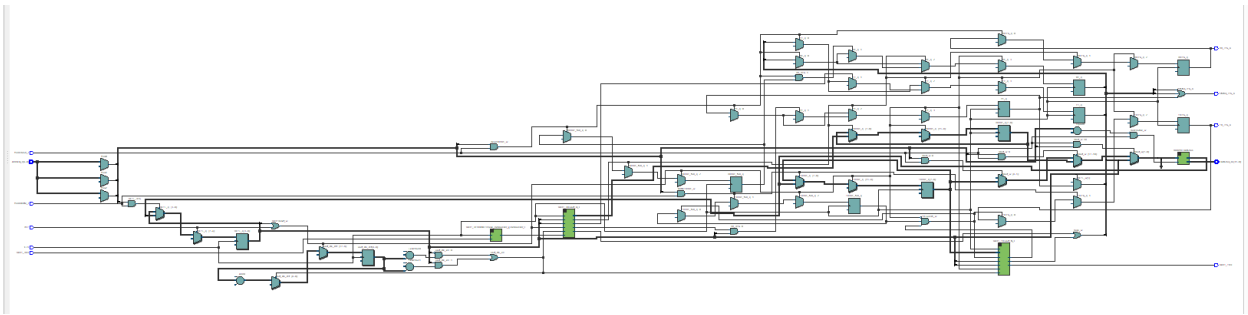


Figure 4: UART — RTL Viewer. Transmit and receive shift registers with their baud-rate counters, parity generation/checking and the status/control registers exposed on the peripheral bus.

3. PPA Characterization

All figures below are taken from Quartus Prime 25.1std.0 compilations targeting the Cyclone V 5CSXFC6D6F31C6 device (Timing Models: Final). On Cyclone V the area metric is Adaptive Logic Modules (ALMs) rather than Cyclone-IV logic elements, and the embedded 9-bit multipliers are realised inside variable-precision DSP blocks.

3.1 Area

Configuration	Logic elements (ALMs)	Registers	I/O pins	Embedded memory bits	Embedded 9-bit multipliers	PLLs
MCU with GPIO	1,903	1,703	70	131,072	4	2
MCU with GPIO and Interrupt Capability	2,017	1,743	70	131,072	4	2
Pipelined MCU with GPIO and Interrupt Capability	—	—	—	—	—	—

Table 1: Area characterization (Quartus Analysis & Synthesis Resource Usage Summary). Combinational ALUT usage is 2,481 and 2,615 respectively.

Adding the interrupt capability, the UART and the divider accelerator costs +114 ALMs (+6.0 %), +134 combinational ALUTs and +40 dedicated registers — a small increment, because the added logic is dominated by narrow control registers (IE, IFG, IR, Type) and by FSM state rather than by datapath width. The I/O pin count is unchanged at 70, and the block-memory, DSP and PLL usage are identical in both configurations, because the ITCM/DTCM sizes, the 16-bit multiplier and the two clock domains (MCLK and DIVCLK) are common to both builds.

The maximum fan-out node differs between the two builds: in the GPIO configuration it is the global reset `rst_w` with 1,474 loads, while in the full configuration it becomes the MCLK net with 1,386 loads. Both are expected for a design in which every architectural register is clocked from one net and cleared from another; the average fan-out is essentially constant (4.71 versus 4.72).

Analysis & Synthesis Resource Usage Summary		
<<Filter>>		
	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1903
2		
3	► Combinational ALUT usage for logic	2481
4		
5	Dedicated logic registers	1703
6		
7	I/O pins	70
8	Total MLAB memory bits	0
9	Total block memory bits	131072
10		
11	Total DSP Blocks	4
12		
13	► Total PLLs	2
14		
15	Maximum fan-out node	rst_w
16	Maximum fan-out	1474
17	Total fan-out	20742
18	Average fan-out	4.71

Figure 5: MCU with GPIO — Analysis & Synthesis resource summary (1,903 ALMs, 1,703 registers, 4 DSP, 2 PLLs).

Compilation Report - work X		RISCV_simple.sdc X
Table of Contents		Analysis & Synthesis Resource Usage Summary
<<Filter>>		
	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	2017
2		
3	► Combinational ALUT usage for logic	2615
4		
5	Dedicated logic registers	1743
6		
7	I/O pins	70
8	Total MLAB memory bits	0
9	Total block memory bits	131072
10		
11	Total DSP Blocks	4
12		
13	► Total PLLs	2
14		
15	Maximum fan-out node	MCU...clk
16	Maximum fan-out	1386
17	Total fan-out	21589
18	Average fan-out	4.72

Figure 6: MCU with GPIO and Interrupt Capability — resource summary (2,017 ALMs, 1,743 registers, 4 DSP, 2 PLLs).

3.2 Performance

Configuration	f_max	Critical path (slowest sub-module and why)	f_MCLK = f_sysclk
MCU with GPIO	28.37 MHz	The same ITCM → IDECODE register-file read → divider operand synchroniser path; the peripheral set differs but the limiting sub-module is identical, which is why the two configurations close at almost the same frequency.	25 MHz

Configuration	f_max	Critical path (slowest sub-module and why)	f_MCLK = f_sysclk
MCU with GPIO and Interrupt Capability	29.09 MHz	ITCM (altsyncram) instruction output → instruction-bit decode (Instruction_ID_w[22]~3) → the IDECODE register-file read multiplexer chain (Mux48~2 → Mux48~4 → Mux48~10) → the divider operand register Bin_q[15] inside Sync:Synchronizer. The slowest sub-module is IDECODE: the fetched instruction must select a register-file source through three cascaded 48-input multiplexers and the read data must still reach the CDC input register of the division accelerator within one MCLK period. Interconnect dominates (IC 1.624 ns on Mux48~2, 0.906 ns on Bin_q[15]) rather than cell delay.	25 MHz
Pipelined MCU with GPIO and Interrupt Capability	—	Expected to migrate to the IDECODE register-file write path, as measured in LAB5.	—

Table 2: Performance characterization (Slow 1100 mV 85 °C model). f_MCLK is the frequency actually programmed into the PLL and used by the system; it is set to 25 MHz, safely below the Fmax of both implemented configurations.

The Fmax report lists four clocks. MCLK — the PLL output that clocks the CPU and the peripherals — closes at 29.09 MHz and is therefore the system limit: $f_MCLK = f_sysclk \leq 29.09$ MHz. DIVCLK, the separate PLL output feeding the divider accelerator, closes at 139.55 MHz, so the accelerator is roughly 4.8 times clear of the bottleneck and the CDC wrapper correctly isolates the two domains. The 1,121.08 MHz figure reported on clk is the raw 50 MHz input port before the PLL and covers only the trivial input-to-PLL path, restricted to 717.36 MHz by the device minimum period, while altera_reserved_tck (94.15 MHz) belongs to the JTAG / Signal-Tap infrastructure and is not part of the user design.

Compared with the 34.99 MHz achieved by the bare single-cycle core in LAB5, the MCU loses roughly 17 % of its clock frequency. The technology-map view of the worst-case path shows where it went: the path starts at the ITCM output, passes through the instruction-bit decode into the IDECODE register-file read multiplexer chain, and ends at the operand register Bin_q[15] of the divider synchroniser. In the bare core the register-file read fed only the ALU; in the MCU the same read data must additionally reach the CDC input register of the division accelerator in the same cycle, which lengthens the fan-out and the routing of that net. Interconnect, not cell delay, dominates every hop of the path.

Between the two MCU configurations the difference is small: 28.37 MHz with GPIO alone against 29.09 MHz with the full peripheral set, i.e. 2.5 % in favour of the larger design. Since additional logic cannot physically shorten a critical path, this inversion is placement-and-routing variation between two independent fitter runs rather than an architectural effect. The correct reading is that the interrupt controller, the UART and the divider are effectively free in clock frequency: they add sources to the bus read-back multiplexer and insert the Type/vector multiplexer, but the limiting path is the same ITCM → register-file read → divider-synchroniser path in both builds. The DIVCLK domain closes at a similar value in both compilations (143.12 MHz and 139.55 MHz), confirming that the accelerator is never the bottleneck.

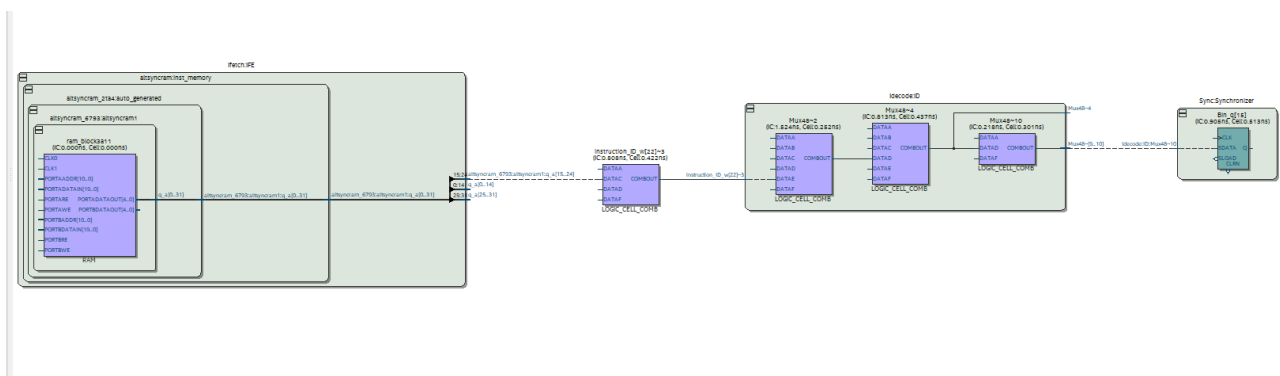


Figure 7: Critical-path technology map. ITCM (altsyncram) → Instruction_ID_w[22]~3 → the IDECODE multiplexer chain Mux48~2 / Mux48~4 / Mux48~10 → Bin_q[15] in Sync:Synchronizer. Per-hop delays are annotated as IC (interconnect) / Cell.

The system is clocked well inside this limit: the PLL is programmed to produce $f_{\text{MCLK}} = f_{\text{sysclk}} = 25 \text{ MHz}$, which leaves roughly 13 % margin over the worst-case 28.37 MHz corner and keeps the design robust across the slow and fast timing models.

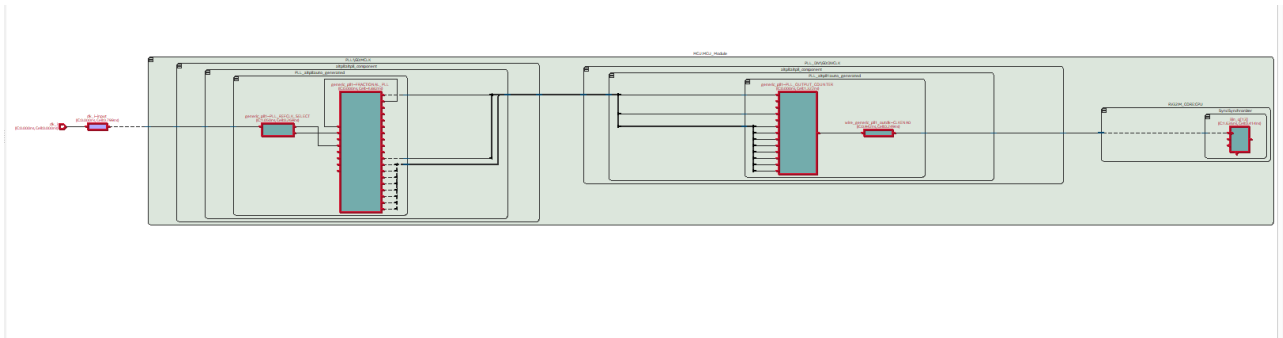


Figure 8: Technology-map view of the clock-generation network. The 50 MHz board clock clk_i enters the fractional PLL $\text{PLL}:\backslash\text{G0}:\text{MCLK}$, whose output counter produces MCLK; the second PLL $\text{PLL}:\backslash\text{G0}:\text{DIVCLK}$ produces the divider-accelerator clock, which re-enters the core through the SyncSynchronizer instance inside $\text{RV32IM_CORE}:\text{CPU}$. This confirms that the two clock domains are physically separated and crossed only through the synchroniser.

Slow 1100mV 85C Model Fmax Summary			
<<Filter>>			
	Fmax	Restricted Fmax	Clock Name
1	28.37 MHz	28.37 MHz	MCU_Module \G0:MCLK a...OUTPUT_COUNTER divclk
2	95.56 MHz	95.56 MHz	altera_reserved_tck
3	143.12 MHz	143.12 MHz	MCU_Module \G0:DIVCLK...OUTPUT_COUNTER divclk
4	1118.57 MHz	717.36 MHz	clk

Figure 9: MCU with GPIO — Slow 1100 mV 85 °C Fmax summary. MCLK = 28.37 MHz (system limit); DIVCLK = 143.12 MHz.

Slow 1100mV 85C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	29.09 MHz	29.09 MHz	MCU_Module \G0:MCLK a...OUTPUT_COUNTER divclk	
2	94.15 MHz	94.15 MHz	altera_reserved_tck	
3	139.55 MHz	139.55 MHz	MCU_Module \G0:DIVCLK...OUTPUT_COUNTER divclk	
4	1121.08 MHz	717.36 MHz	clk	limit due to minimum...d rest

Figure 10: MCU with GPIO and Interrupt Capability — Slow 1100 mV 85 °C Fmax summary. MCLK = 29.09 MHz (system limit); DIVCLK = 139.55 MHz.

3.3 Power

Configuration	Total power	Static power	Dynamic power	I/O power
MCU with GPIO	453.88 mW	415.34 mW	25.14 mW	13.40 mW
MCU with GPIO and Interrupt Capability	455.09 mW	415.35 mW	26.36 mW	13.39 mW
Pipelined MCU with GPIO and Interrupt Capability	—	—	—	—

Table 3: Power characterization (PowerPlay, Final timing models). Estimation confidence is Low because the .vcd toggle data is partial, so the absolute numbers are indicative and only the relative comparison is meaningful.

Static power (≈ 415.3 mW) is identical in both configurations, as expected: it depends on the device, the process corner and the junction temperature, not on the utilised logic. The whole difference therefore appears in the dynamic term, which grows from 25.14 mW to 26.36 mW (+4.9 %) — consistent with the +40 registers and +114 ALMs that toggle every MCLK edge. I/O power is flat (13.40 $\rightarrow$ 13.39 mW) because the pin count is unchanged at 70 and the added peripherals drive internal nodes rather than new pins.

Total Thermal Power Dissipation	453.88 mW
Core Dynamic Thermal Power Dissipation	25.14 mW
Core Static Thermal Power Dissipation	415.34 mW
I/O Thermal Power Dissipation	13.40 mW

Figure 11: MCU with GPIO — PowerPlay summary: 453.88 mW total (415.34 static / 25.14 dynamic / 13.40 I/O).

Total Thermal Power Dissipation	455.09 mW
Core Dynamic Thermal Power Dissipation	26.36 mW
Core Static Thermal Power Dissipation	415.35 mW
I/O Thermal Power Dissipation	13.39 mW

Figure 12: MCU with GPIO and Interrupt Capability — PowerPlay summary: 455.09 mW total (415.35 static / 26.36 dynamic / 13.39 I/O).

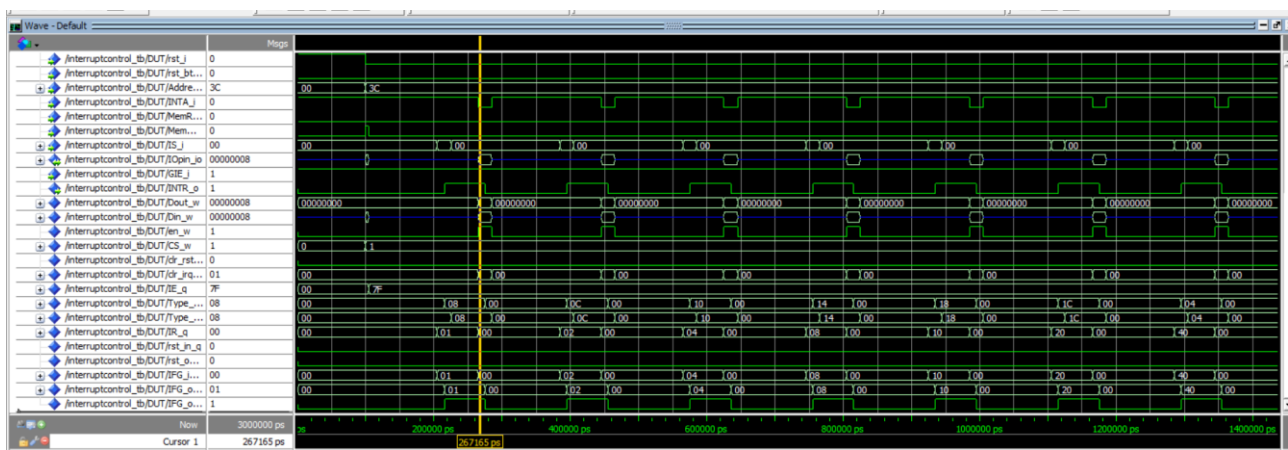


Figure 15: Interrupt Controller unit test bench — IFG/IR/Type_out sequence over seven consecutive requests with the INTR/INTA hand-shake.

4.2 System-level verification (tb_mcu)

With the peripherals proven, the structural MCU was simulated running the benchmark program. The top-level waveform shows the PC stepping through the program loop (0x1C → 0x20 → 0x24 → 0x28 → 0x2C → 0x3C → 0x40 → 0x50 → 0x54 → 0x64 and back) with the matching instruction words fetched from the ITCM, GIE_o and INTA_o asserted, and the HEX/LEDR outputs holding their programmed values. The reset push-buttons KEY0–KEY3 are released (logic 1) and SW = 0x3FF is presented to the GPIO.

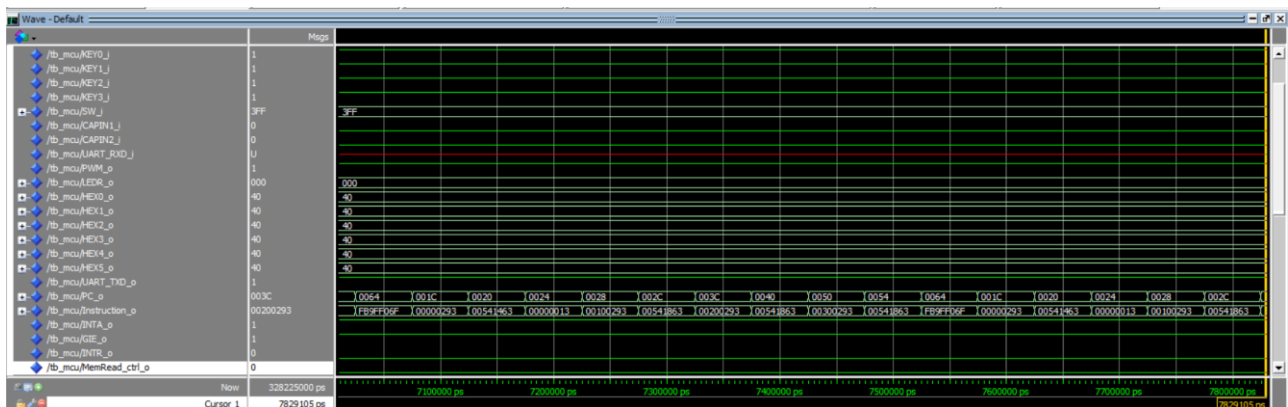


Figure 16: MCU top level — `tb_mcu` waveform showing the PC/instruction stream, the interrupt hand-shake signals and the board-level outputs.

The CPU view confirms the single-cycle behaviour inside the assembled system: one instruction retires per MCLK period, RegWrite_ctrl_o pulses on every arithmetic instruction, alu_res_o and write_data_o present the value in the same cycle in which dtcm_addr_o is driven, and the free-running mclk_cnt_o increments monotonically. Both clock domains (clk_i and divclk_i) are visible and running at their intended ratio.

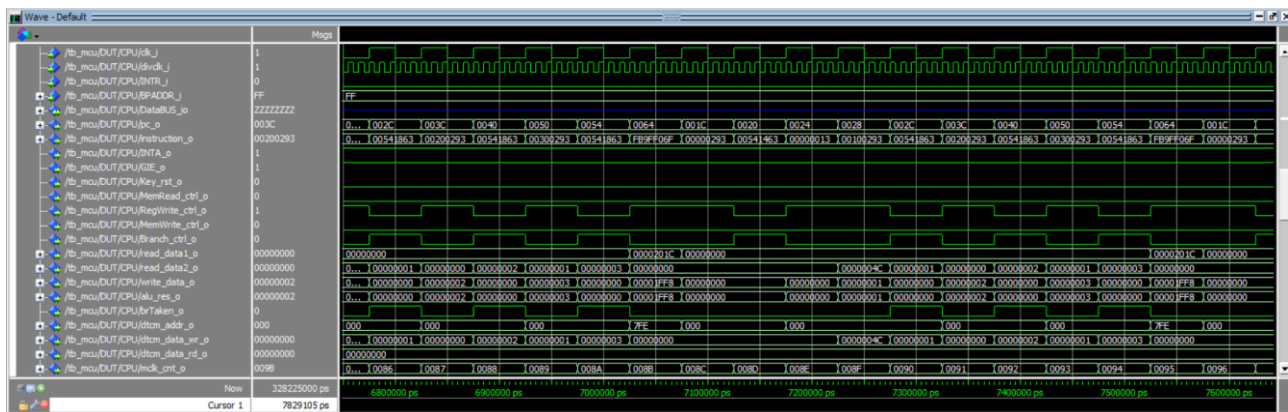


Figure 17: MCU — internal CPU signals (PC, instruction, control strobes, ALU/DTCM datapath and the MCLK counter) during the benchmark run.

The peripheral instances were observed at the same simulation time (cursor at 109.3 ns) so that the bus transactions can be correlated across blocks. In GPIO_UNIT the address bus walks through 0x1A, 0x20, 0x26,

Figure 18: MCU — GPIO_UNIT instance: address decoding, chip-select generation and port registers during system operation.

Figure 19: MCU — IC_UNIT instance: IE/Type registers and the INTR/INTA/GIE signalling seen from inside the assembled MCU.

Taken together, the unit-level and system-level simulations show that every peripheral behaves identically inside the MCU as it did standalone, that the shared tri-state bus is driven by exactly one source at a time, and that the interrupt hand-shake operates correctly against the ISR FSM in the control unit.

0x0044, 0x0048) with the matching instruction words, and the free-running MCLK counter increments monotonically (0xF54E → 0xF55D), confirming that the core runs from the PLL-generated MCLK without stalling.

The peripheral outputs are captured on the same trigger: HEX0 = 0x78 and HEX1 = 0x02 hold the values written by the benchmark through the GPIO port registers, HEX2–HEX5 remain at their reset pattern 0x40, LEDR is cleared, PWM_o is low in the captured window and UART_TXD idles high as expected for an idle line. The reset net rst_w is de-asserted and KEY0–KEY3 read as released, so the capture represents normal operation rather than a reset condition. These values match the ModelSim run of the same benchmark, which confirms that the MCU behaves identically in simulation and on the board.



Figure 21: On-board Signal-Tap capture of the MCU top-level ports — PC and instruction stream, MCLK counter, push-buttons, UART lines, PWM and the HEX / LEDR outputs.

5. Conclusions

A complete structural RV32IM micro-controller was implemented on the DE10-Standard board, combining the single-cycle core of LAB5 with a memory-mapped peripheral subsystem (GPIO with SSD decoders, Basic Timer with PWM and capture, Interrupt Controller, UART and a restoring-divider accelerator on a separate clock domain).

The PPA comparison between the two implemented configurations shows that interrupt capability, serial communication and the division accelerator are inexpensive in area (+6.0 % ALMs, +2.3 % registers, with identical memory, DSP and PLL usage) and almost free in power (+1.21 mW total, entirely in the dynamic term). They are also effectively free in clock frequency: 28.37 MHz with GPIO alone against 29.09 MHz with the full peripheral set, a 2.5 % spread that reflects fitter variation rather than an architectural difference. The real performance cost is the step from a bare CPU to a micro-controller: 29.09 MHz against 34.99 MHz for the LAB5 core. The technology-map analysis locates the worst-case path in the IDECODE register-file read multiplexer chain, running from the ITCM output to the operand register of the divider synchroniser, so the added fan-out of the register-file read — not the peripheral bus itself — is what limits the clock. The divider datapath, in its own 139.55 MHz DIVCLK domain, is comfortably clear. The system is operated at $f_{\text{MCLK}} = f_{\text{sysclk}} = 25 \text{ MHz}$, inside the worst-case F_{max} of both configurations.

Functional correctness was established bottom-up: each peripheral was verified standalone against its own test bench, then re-verified in place inside the assembled MCU running the benchmark program, and finally confirmed on the board with Signal-Tap, which reproduced the same PC / instruction stream and the same peripheral outputs as the RTL simulation.

Remaining work. The GPIO-only F_{max} report and the full PPA set for the pipelined configuration still have to be captured; once available they complete the three rows of Tables 1–3 and allow the same single-cycle-versus-pipelined trade-off analysis performed in LAB5 to be repeated at MCU level.